THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A block is pulled with 4 N but does not move. The friction acting on it is: 1

(a) 8 N **(b)** 2 N **(c)** 4 N **(d)** 0 N

2. A glass rod is rubbed with silk and brought near a suspended charged plastic rod, which is repelled. The plastic rod has: 1

(a) magnetism **(b)** negative charge **(c)** positive charge **(d)** no charge

3. Which of the following is NOT an effect of force? 1

(a) change in speed **(b)** change in direction

(c) change in shape **(d)** change in colour

4. A cart is pushed with 20 N and friction is 12 N. The net force is: 1

(a) 12 N **(b)** 8 N **(c)** 32 N **(d)** 20 N

5. A charged balloon attracts a neutral wall because: 1

(a) the balloon is heavy **(b)** the wall is a magnet

(c) of gravity only **(d)** it induces opposite charge near it

6. A spring balance shows 3 N for an object. Its mass is about (taking 10 N for 1 kg): 1

(a) 30 kg **(b)** 0.3 kg **(c)** 0.03 kg **(d)** 3 kg

7. Two persons pull a rope with 100 N each in opposite directions. The net force on the rope is: 1

(a) 0 N **(b)** 100 N **(c)** 50 N **(d)** 200 N

8. An electroscope's leaves collapse when a charged rod is touched by a finger because: 1

(a) leaves become heavy **(b)** air enters

(c) a magnet acts **(d)** charge flows to earth

9. A magnet is cut into two pieces. Each piece will: 1

(a) be a south pole only **(b)** lose magnetism

(c) be a north pole only **(d)** have two poles

10. A ball on a table has weight 5 N. The table pushes it with: 1

(a) 5 N upward **(b)** 5 N downward **(c)** 10 N **(d)** 0 N

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A trolley is pushed with 15 N and the friction is 6 N. Find the net force and say in which direction it acts. 2

12. Why do athletes use spikes on their shoes? 2

13. Why does rubbing hands together warm them? 2

14. Why is more force needed to start a heavy almirah moving than to keep it moving? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A box of mass 10 kg is on a floor. A spring balance shows 25 N just before it starts moving and 18 N while sliding. Explain the readings. 3

16. Explain why the weight of an astronaut changes on the Moon but not the mass. 3

17. A bar magnet is dipped in iron filings. Where do most filings stick and why? 3

OR

Explain how a force of 20 N and a force of 12 N can produce both a net force of 32 N and one of 8 N on the same body. 3

18. How do you show that friction depends on the load? Describe a method. 3

OR

Why does a falling leaf fall slower than a stone? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student pulls a block with a spring balance on three surfaces: polished table (2 N), wooden floor (5 N) and carpet (9 N). Explain the observations and draw a conclusion. How will the readings change if a 1 kg mass is added? 5

OR

Discuss the effect of air resistance on a falling feather, a stone and a parachutist. 5

20. Explain, with reasons, which type of friction acts in: 5
(a) a car just starting
(b) a box sliding
(c) a rolling ball
(d) a parked scooter on a slope.

OR

Describe how magnets interact with iron, with each other, and how a compass works. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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